ABET = 85975

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ChE 311 Midterm Exam II, Spring 2019
Closed book and closed notes. Only provided tables are allowed.
50 minutes, 100 points

1. (5 points) Which of the following statements is/are true?

(A) Eddy viscosity of Boussinesq is a property of the fluid.

(B) Eddy viscosity for a flow system is constant everywhere.

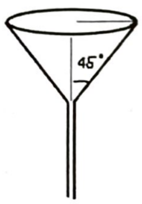
(C) Eddy viscosity of Boussinesq gives a first-order approximation for Reynolds Stress.

(D) None of above.

Answer : C

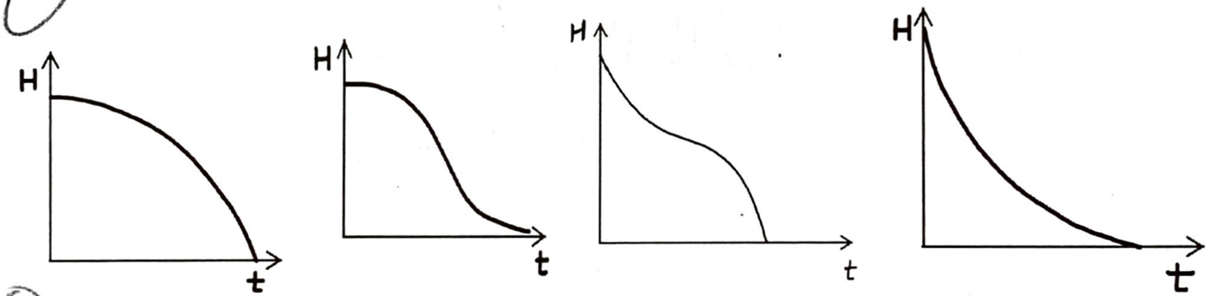
$-\mu^{(t)} \frac{\partial U_x}{\partial y}$

2. (5 points) Which of the following curve represents the draining curve of a conical tank?



Answer: A

(A) (B) (C) (D)



3. (5 points) Estimate the order of the Reynolds number of a swimmer who swims a kilometer in 20 minutes. Choose the closest answer.

(A) 0.01 (B) 1 (C) 100 (D) 100,000

Answer: D

$R_e = \frac{\rho \langle v \rangle l}{\mu}$

4. (5 points) For flow in a circular tube, the coordinate was set up as z was the flow direction and r was the radical direction. Please write down the $\overline{\tau}_{rz}^{(t)}$ by its definition in terms of velocity fluctuation?

$$\overline{\tau}_{rz}^{(t)} = \rho \overline{v_r' v_z'}$$

5. (5 points) For above flow in a circular tube, please write down $\overline{\tau}_{rz}^{(t)}$ in terms of wall shear stress (τ_0) and the radius of the tube (R).

$$\overline{\tau}_{rz}^{(t)} = \tau_0 (1 - y/R)$$

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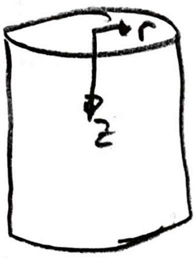
6. (30 points) **Turbulent flow in a circular tube.**

Please determine the ratio of $\mu^{(t)}/\mu$ at y = R/3 for water flowing at a steady rate in a long, sooth, circular tube under the following conditions.



$y = R/3$

Tube radius, R = 10 cm

$\frac{y v_*}{\nu} = y/\nu \sqrt{\tau_0}$

Wall shear stress, τ_0 = 0.2 Pa

Density of the fluid, ρ = 1000 kg/m3

Viscosity of the fluid, μ = (cp) → Memorize !!!

$v_z = f(r)$

For viscous sublayer, $\frac{\overline{v}_x}{v_*} = \frac{y v_*}{\nu}\left[1 - \frac{1}{2}\left(\frac{\nu}{v_* B}\right)\left(\frac{y v_*}{\nu}\right) - \frac{1}{4}\left(\frac{y v_*}{14.5\nu}\right)^3 + \cdots\right] \quad 0 < \frac{y v_*}{\nu} < 5$

For inertial sublayer, $\frac{\overline{v}_x}{v_*} = 2.5 \ln\left(\frac{y v_*}{\nu}\right) + 5.5 \quad \frac{y v_*}{\nu} > 30$

$$\frac{\mu^{(t)}}{\mu} = ? \Rightarrow \underbrace{\overline{\overline{\tau}}_{rz}}_{①} = \overline{\overline{\tau}}^{(v)} + \overline{\overline{\tau}}^{(t)}$$

$$\overline{\tau}_{rz} = ?$$

$$0 = -\frac{\Delta P}{l} + \left(\frac{1}{r}\frac{\partial}{\partial r}(r \tau_{rz}\right)$$

$$\frac{\partial}{\partial r}(r \tau_{rz}) = \frac{\Delta P}{l} \cdot r$$

$$\int \partial (r \tau_{rz}) = \int \frac{\Delta P}{l} \cdot r \, dr$$

$$r \tau_{rz} = \frac{\Delta P}{l}\frac{r^2}{2} + C_0$$

$$\tau_{rz} = \frac{\Delta P}{l}\frac{r}{2} + \frac{C_0}{r}$$

BC $r = 0$ $\tau_{rz} \sim$ finite

$$\tau_{rz} = \frac{\Delta P}{l}\frac{r}{2}$$

$$\Rightarrow \overline{\overline{\tau}}_{rz} = \tau_0 (1 - y/R)$$

$$\underline{\underline{\bar{\tau}}}_{rz} = \underline{\underline{\bar{\tau}}}_{rz}^{(t)} + \underline{\underline{\bar{\tau}}}^{(v)} \qquad \frac{\mu^{(t)}}{\mu}$$

$$\underline{\underline{\bar{\tau}}}_{rz} = -\mu^{t} \frac{\partial \bar{v}_z}{\partial r} - \mu \frac{\partial \bar{v}_z}{\partial r}$$

$$\left(\mu^{(t)} \frac{\partial \bar{v}_z}{\partial r} = -\mu \frac{\partial \bar{v}_z}{\partial r} - \underline{\underline{\bar{\tau}}}_{rz} \right) \frac{1}{\mu} \cdot \frac{1}{\frac{\partial \bar{v}_z}{\partial r}}$$

$$\frac{\mu^{(t)}}{\mu} = \frac{-1}{\mu} \frac{\underline{\underline{\bar{\tau}}}_{rz}}{\partial \bar{v}_z / \partial r} - 1 \qquad \left[\begin{array}{l} y = R - r \\ \partial y = -\partial r \end{array} \right]$$

$$= \frac{-1}{\mu} \frac{\left(\tau_0 (1 - y/R) \right)}{\partial \bar{v}_z / \partial y} - 1 \qquad \frac{\partial}{\partial r} \to \frac{\partial}{\partial y}$$

Break (Non-dimensional)

- $v_* = \sqrt{\frac{\tau_0}{\rho}}$ • $\nu = \frac{\mu}{\rho}$ • $y^+ = \frac{y v_*}{\nu}$ • $v^+ = \frac{\bar{v}}{v_*}$

$$v_*^2 = \frac{\tau_0}{\rho}$$

$$\tau_0 = \rho v_*^2$$

Back

$$= +\frac{1}{\mu} \rho v_*^2 \frac{(1 - y/R)}{\partial \bar{v}_z / \partial y} - 1$$

Note ∂ constant x = const ∂x (Doesn't matter)

$$\frac{x}{y} = \frac{1}{y/x}$$

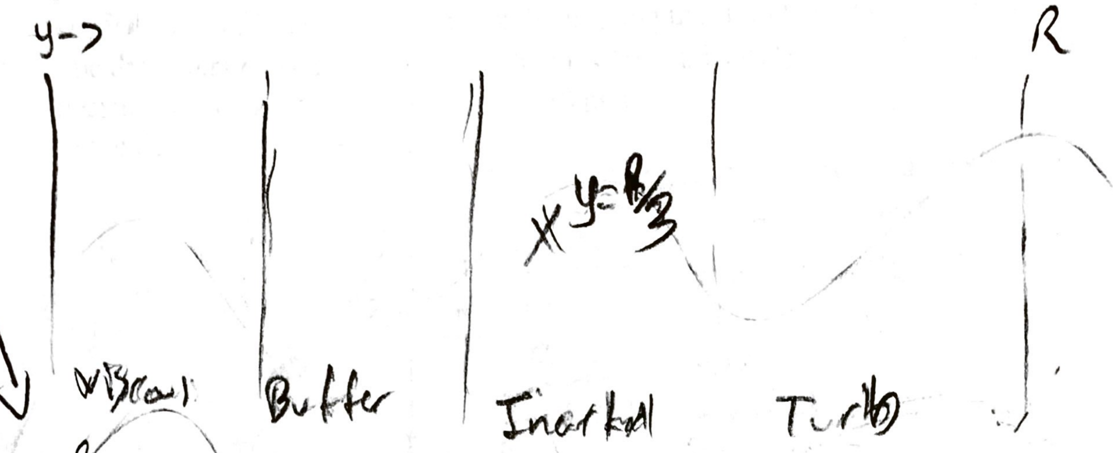
$$= \frac{v_* v_*}{\nu} \frac{(1 - y/R)}{\partial \bar{v}_z / \partial y} - 1 \qquad v^+ = \frac{\bar{v}}{v_*}$$

$$\Rightarrow = \frac{(1 - y/R)}{\frac{\partial (\bar{v}_z / v_*)}{\partial (y v_* / \nu)}} - 1$$

$$\frac{\mu^{(t)}}{\mu} = \frac{(1 - y/R)}{\frac{\partial v^+}{\partial y^+}} - 1$$

$$\frac{\mu(t)}{\mu} = \frac{(1 - y/R)}{\frac{\partial v^+}{\partial y^+}} - 1$$

$$\frac{\partial v^+}{\partial y^+} = ?$$



$y = R/3$ Not close to wall

$$\Rightarrow \underbrace{\frac{\bar{v}_x}{v_*}} = 2.5 \ln\Big(\underbrace{\frac{y v_*}{\nu}}\Big) + 5.5 \Rightarrow \frac{\mu(t)}{\mu} = \frac{(1 - y/R)}{\left(\frac{2.5}{y^+}\right)} - 1$$

$$v^+ = 2.5 \ln(y^+) + 5.5 \qquad = \frac{(1 - \frac{1/3 R}{R})}{\frac{2.5}{(157.04)}} - 1$$

$$\frac{\partial v^+}{\partial y^+} = \frac{2.5}{y^+} \qquad \frac{\mu(t)}{\mu} = 40.9026$$

Actual value $\rightarrow \frac{y v_*}{\nu} \Rightarrow y^+ = ?$ dimensionless

$$\frac{\left(\frac{R}{3}\right)\left(\sqrt{\frac{\tau_0}{\rho}}\right)}{\left(\frac{\mu}{\rho}\right)} = \frac{\left(\frac{0.1\,m}{3}\right)\left(\sqrt{\frac{0.2\,Pa \left|\frac{1\ N/m^2}{1\ Pa}\right|\left|\frac{1\ kg\,m/s^2}{N}\right|}{1000\,kg/m^3}}\right)}{\frac{1\,cp \left|\frac{1\times10^{-4}}{1\,cp}\right|}{1000\,kg/m^3}} = 157.1346 \approx 5$$

Assuming $\rightarrow$ 1cp = ? $1\times10^{-4}\ Pa\cdot s^2$

If 1cp = something else + $y^+ < 5$ then use viscous equation by differentiating it + plugging in above

7. **(25 points) Using a conical funnel to measure the viscosity of a fluid.**
The dimensions of the conical funnel are shown in the figure. The initial liquid level is at Z_1=5cm. The efflux time for completely draining the fluid (t_{efflux}) is 100s. ~~$R_0=Z_0=8cm$~~. The pipe diameter (D) is 2mm. Pipe length (L) is 10 cm. Density of the fluid (ρ) has been measure, which is 1500 kg/m^3. Take g=10 m/s^2.
Please decide the viscosity of the fluid.

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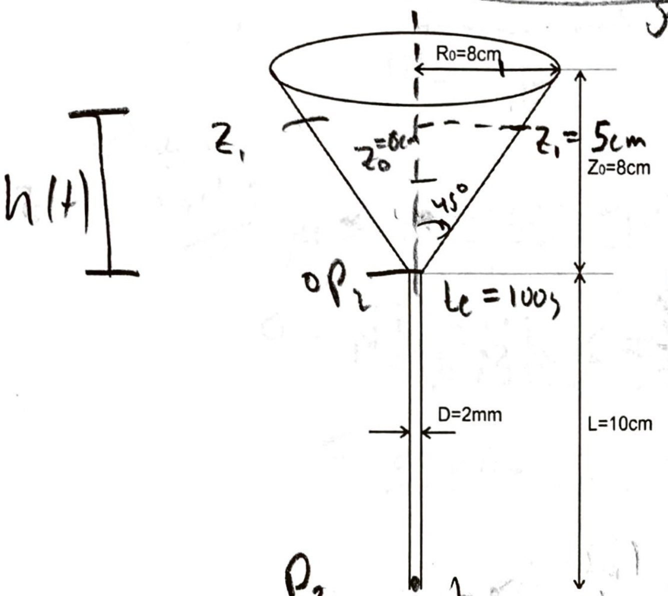
$\rho = 1500\ kg/m^3$ $\quad g = 10\ m/s^2$

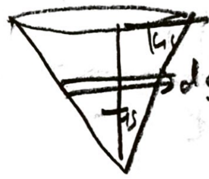


$$\frac{dm_{tot}}{dt} = \cancel{w_1}^{0} - w_2$$

$$\underbrace{\frac{dm_{tot}}{dt}}_{①} = \underbrace{-w_2}_{②}$$

h

2o)



$\Rightarrow \forall = \pi x^2 dy$ (y)

$x = y$ from 45° Δ

① $m_{tot} = \rho \forall = \rho \int_0^h (\pi x^2)\, dy$

$= \rho \int_0^h \pi y^2\, dy$

$= \rho \left(\pi \frac{y^3}{3}\right]_0^{h}$

$= \rho \pi \frac{h^3}{3}$

② $-w_2 = -\rho(Q) = -\rho\left(\frac{\pi \Delta P R_p^4}{8 \mu L}\right)$ ✓ $R_p = \frac{1}{2}D$

$= \frac{-\rho \pi \Delta P \left(\frac{1}{2}D\right)^4}{8 \mu L}$

$= \frac{-\rho \pi \Delta P D^4}{128 \mu L}$

$\Delta P = P_2 - P_3 + \rho g L$

$= (\cancel{P_a} + \rho g h) - \cancel{P_a} + \rho g L$

$= \rho g (h + L)$ $\Rightarrow -w_2 = \frac{-\rho \pi \rho g (h+L) D^4}{128 \mu L}$

$$\Rightarrow \frac{dm_{tot}}{dt} = -\omega_2$$

$$\frac{d}{dt}\left[\rho\pi \frac{h^3}{3}\right] = -\rho \frac{\pi(\rho g(h+L))D^4}{128\mu L}$$

Total diff

$$\cancel{\rho\pi}\frac{1}{3}\, d[h^3] = \frac{-\cancel{\rho\pi}(\rho g(h+L))D^4}{128\mu L}\,dt$$

$$\frac{1}{\cancel{3}}(\cancel{3}h^2\,dh) = -\frac{\rho g D^4}{128\mu L}(h+L)\,dt$$

$$\frac{h^2\,dh}{(h+L)} = -\underbrace{\frac{\rho g D^4}{128\mu L}}_{\lambda}\,dt$$

Break

$$\lambda = \frac{\rho g D^4}{128\mu L}$$

$$H = h + L \Rightarrow dH = dh + 0$$
$$h = H - L$$

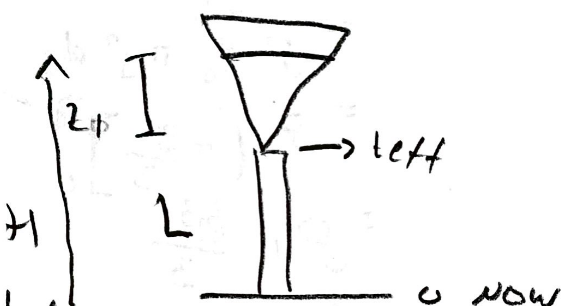
Back

$$\Rightarrow \frac{(H+L)\,dH}{(H-\cancel{L})+\cancel{L}} = -\lambda\,dt$$

Go back to fix

$$\int_{L+z_1}^{L} \frac{(H-L)\,dH}{H} = -\lambda\,dt$$

$$\int \left(1-\frac{L}{H}\right)dH = \int_0^{t_{eff}} -\lambda\,dt$$



$$\left(H - L\ln H\right]_{L+z_1}^{L} = -\lambda t\Big]_0^{t_{eff}}$$

$$\cancel{L} - L\ln(L) - (\cancel{L} + z_1 - L\ln(L+z_1)) = -\lambda t_{eff} - 0$$

$$-z_1 - L\ln L + L\ln(L+z_1) = -\lambda t_{eff}$$

$$-z_1 + L(\ln(L+z_1) - \ln L) = -\lambda t_{eff}$$

$$-z_1 + L\left(\ln\left(\frac{L+z_1}{L}\right)\right) = -\lambda t_{eff}$$

$$-z_1 + L\ln\left(1+\frac{z_1}{L}\right) = -\lambda t_{eff}$$

$$-z_1 + L\ln\left(1+\frac{z_1}{L}\right) = -\lambda t_{eff}$$

Solving for μ

$$-z_1 + L\ln\left(1+\frac{z_1}{L}\right) = -\frac{\rho g D^4}{128\mu L} t_{eff}$$

$$\mu\left(-z_1 + L\ln\left(1+\frac{z_1}{L}\right)\right) = -\frac{\rho g D^4}{128 L} t_{eff}$$

$$\mu = \frac{-\rho g D^4}{128\ L} t_{eff} \cdot \left(\frac{1}{-z_1 + L\ln\left(1+\frac{z_1}{L}\right)}\right)$$

Plug in vales

$$\mu = -\frac{(1100 kg/m^3)(10 m/s^2)(0.002m)^4}{128 \cdot 0.01m} \cdot 1008\left(\frac{1}{-0.05m + 0.01m\ln\left(1+\frac{0.05}{0.01}\right)}\right)$$

$$= -1.875\times10^{-5} \frac{kgm^2/s}{m/1} \cdot \left(-31.1697 \frac{1}{m}\right)$$

$$= 5.844\times10^{-4} \frac{kg}{s}.$$ ×